

$$\hat{\theta} = \min \left(\frac{1 - p^2 - 2c}{2p(1 - 2p)}, \frac{1}{2} \right)$$

We have the constraint:

$$q = \frac{1 - p^2 - 2c}{p(1 - 2p)} \in [0, 1]$$

And types that do disclose are $\theta_0 > 2\hat{\theta} = q$ Now we are maximizing:

$$P = c(1 - 2\hat{\theta}) = c \left(1 - \frac{1 - p^2 - 2c}{p(1 - 2p)} \right)$$

Thus, the DMV solves the constrained optimization problem:

$$\max_{c, p} c \left(1 - \frac{1 - p^2 - 2c}{p(1 - 2p)} \right)$$

subject to

$$0 \leq \frac{1 - p^2 - 2c}{p(1 - 2p)} \leq 1, \quad p \in [0, 1]$$

We first consider $p > 1/2$. Clearly this is a concave problem in c for any fixed $p = \bar{p}$ as

$$c \left(1 - \frac{\bar{p}^2 - 2c}{\bar{p}(1 - 2\bar{p})} \right) = c \left(1 + \frac{\bar{p}^2}{\bar{p}(2\bar{p} - 1)} - \frac{2c}{\bar{p}(2\bar{p} - 1)} \right)$$

where $\frac{1}{\bar{p}(2\bar{p}-1)} > 0$ so it is a downward opening sloping quadratic in c .

Thus, the FOC (for fixed p) is

$$[c] : \quad 1 - \frac{1 - p^2 - 2c}{p(1 - 2p)} + c \left(\frac{2}{p(1 - 2p)} \right) = 0$$

$$p(1 - 2p) - (1 - p^2) + 4c = 0$$

$$c = \frac{1 - p + p^2}{4}$$

Then profit is

$$P^*(p) = \left(\frac{1 - p + p^2}{4} \right) \left(1 - \frac{1 - p^2 - 2 \cdot \left(\frac{1 - p + p^2}{4} \right)}{p(1 - 2p)} \right)$$

And constraint is

$$\frac{1 - p^2 - 2 \cdot \left(\frac{1-p+p^2}{4} \right)}{p(1-2p)} \geq 0$$

$$\frac{1 - p^2 - \frac{1}{2}(1 - p + p^2)}{p(1-2p)} = \frac{2 - 2p^2 - (1 - p + p^2)}{2p(1-2p)} = \frac{1 + p - 3p^2}{2p(1-2p)} \geq 0$$

Thus, $1 + p - 3p^2 \leq 0 \implies p \geq p_{min} := \frac{1+\sqrt{13}}{6} \approx 0.7676$

For $p > p_{min}$ then there is an interior solution.

At $p = p_{min}$ all types except lemons ($\theta = 0$) will disclose.

We can simplify profit:

$$\begin{aligned} P^*(p) &= \left(\frac{1 - p + p^2}{4} \right) \left(1 - \frac{1 + p - 3p^2}{2p(1-2p)} \right) \\ &= \left(\frac{1 - p + p^2}{4} \right) \left(\frac{p - 1 - p^2}{2p(1-2p)} \right) \\ &= \frac{(1 - p + p^2)^2}{8p(2p - 1)} \end{aligned}$$

As expected, when $p = 1$ we have $P^*(1) = 1/8$ which is the baseline with no signal manipulation.

At $p = p_{min}$ the profit is:

$$P^* \left(\frac{1 + \sqrt{13}}{6} \right) = \frac{11 - \sqrt{13}}{36} \approx 0.2054$$

Note that this already greatly exceeds the DMV's baseline profit of $P^*(1) = 0.125$ under a perfect signal. Now if $1/2 < p < p_{min}$ then the DMV chooses c so that $\hat{\theta} = 0$ is binding, so we have a corner solution, ie. all types except the lemons ($\theta = 0$) disclose to maximize profit.

Thus we have

$$\begin{aligned} q &= \frac{1 - p^2 - 2c}{p(1-2p)} = 0 \\ c &= \frac{1 - p^2}{2} \end{aligned}$$

And we get

$$P^*(p) = c \cdot (1 - 0) = \frac{1 - p^2}{2}$$

Now let us consider $p < 1/2$. In this case, $P_p(c)$ is an upward facing parabola with minimum at $c = \frac{1-p+p^2}{4}$. And the constraint is $0 \leq \frac{1-p^2-2c}{p(1-2p)} \leq 1$ and $1 - 2p > 0$ so rewrite this as

$$0 \leq 1 - p^2 - 2c \leq p - 2p^2$$

which rearranges to

$$\frac{1 - p + p^2}{2} \leq c \leq \frac{1 - p^2}{2}$$

But note that

$$\frac{1 - p + p^2}{4} \leq \frac{1 - p + p^2}{2} < \frac{1 - p^2}{2}$$

and thus clearly P_p is maximized at $c = \frac{1-p^2}{2}$ as P_p is increasing on $(\frac{1-p+p^2}{4}, \infty)$

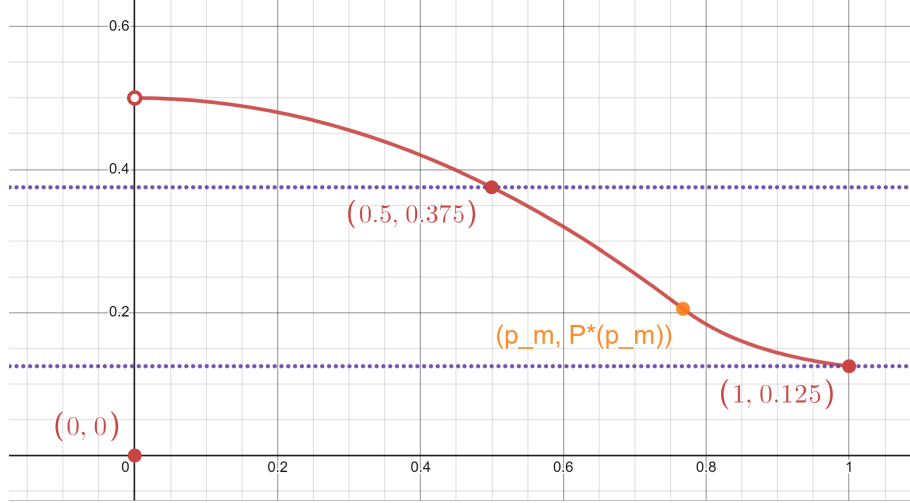
Plugging this in we get

$$P^*(p) = \frac{1 - p^2}{2} \left(1 - \frac{1 - p^2 - 2 \cdot \frac{1-p+p^2}{2}}{p(1-2p)} \right) = \frac{1 - p^2}{2}$$

And so for $p < 1/2$ the DMV again charges sufficiently low cost to induce all types except lemons ($\theta = 0$) to disclose.

We can now summarize the optimal DMV profit $P^*(p)$ for all $p \in [0, 1]$ as follows:

$$P^*(p) = \begin{cases} \frac{(1 - p + p^2)^2}{8p(2p - 1)}, & p \geq p_{min} \\ \frac{1 - p^2}{2}, & 0 < p < p_{min} \\ 0, & p = 0 \end{cases}$$



The DMV's profit is increasing with decreasing p , ie. by lowering the informativeness of the signal the DMV can make more overall. In fact the DMV can achieve any profit less than $(1 - 0^2)/2 = 1/2$ by choosing sufficiently low values of p . However, if the signal is completely uninformative ($p = 0$) then the DMV makes no profit as no seller will purchase the certificate. Recall that in the baseline model with perfect information the DMV made profit $P = 1/8$.

Thus the DMV can achieve almost **quadruple** the profit compared to the perfect signal in the baseline model. This is a surprising and compelling result. By degrading the quality of the signal, the DMV can actually increase its overall profit, even by charging the same amount c per seller for any type. By introducing uncertainty into the signal, the DMV can actually encourage lower θ types to “disclose,” because there is some possibility that their revealed s is higher than their true θ when the DMV gives the fake signal. Indeed, in the baseline model, only the sellers with types $\theta > 1/2$ disclose. And if $s \sim U(0, 1)$ then $E(s) = 1/2$, so a seller of type $\theta < 1/2$ has an incentive to disclose with this possibility of misrepresenting its θ to a higher type.

So by degrading the quality of the signal this way, the DMV incentivizes the lower type sellers to disclose and can charge more per seller. This continues until the DMV can incentivize all types except lemons ($\theta = 0$) to disclose, which maximizes profit.

We also have the optimal cost that the DMV charges,

$$c^*(p) = \begin{cases} \frac{1-p+p^2}{4}, & p \geq p_{min} \\ \frac{1-p^2}{2}, & 0 < p < p_{min} \\ 0, & p = 0 \end{cases}$$

Note that c^* is increasing on $(p_{min}, 1)$ and decreasing on $(0, p_{min})$.

We see that for $p > p_{min}$, there is a tradeoff between lowering the cost per seller to attract more sellers (lower types) to disclose, and the profit made per seller that discloses. As p decreases and the signal becomes less informative, lower types become willing to disclose, but the DMV continues to charge lower costs. At the threshold $p = p_{min}$, the DMV is able to capture the entire market at optimum, with all types except lemons disclosing. When the DMV further decreases p , it can then begin to raise the cost again while still getting all non-lemons to disclose.

Perhaps the most surprising result is that even for $p < 1/2$, the DMV continues to make more profit by further decreasing the quality of the signal, and charging even more per seller. For such p , the DMV has $P^*(p) = c^*(p) = \frac{1-p^2}{2}$ and all types except lemons ($\theta = 0$) disclose. This can be understood as follows: When the DMV continues to degrade the quality of its signal, this increases the lower types ($\theta < 1/2$) willingness to pay, because the probability of getting a fake signal, $1 - p$, increases, and the fake signal is on average better than the true θ being revealed. Thus, the lower types are willing to pay more to receive the certificate. The higher types, on the other hand, suffer from the lower signal quality, but have no choice and still must disclose, because the alternative is to be grouped in with the undisclosed types, which are the very low types, ie. lemons. So they are willing to pay the higher cost of disclosing, even as the signal quality continues to degrade. This is the reason why the car dealers still employ a cutoff strategy for any $p > 0$ signal structure.

At optimum, the DMV charges cost c equal to the lemons' reservation, just high enough to make the lemons indifferent and not disclose. Then the DMV is able to incentivize all higher types to disclose.

We observe, however, that by lowering the signal quality, overall welfare is clearly worse.